

Question			Marking details	Marks available					Prac
				AO1	AO2	AO3	Total	Maths	
2	(a)	(i) (ii)	Mass of a molecule [accept: atom or particle] [of the gas] Mean [accept: average] square velocity / speed [of a molecule of the gas]	1 1			2		
	(b)		N_A , [Avogadro's number is] the number of molecules per mole [or in a mole] [of gas] (1) There are n moles [of the gas] (1) So the total number of molecules [in the gas] = $n N_A$ (1)	3			3		
	(c)		Realising that $KE = N_{(A)} \frac{1}{2} \overline{mc^2}$ (1) Substitution of $pV = nRT$ (1) Convincing algebra, e.g handling $\frac{2}{3} \frac{1}{3} \frac{3}{2}$ (1) Explanation that $n = 1$ [NB. if M used need to see molar mass when $n = 1$] (1) Alternative: $\frac{1}{2} \overline{mc^2} = \frac{3}{2} kT$ (1) Multiplication by $N \rightarrow \frac{1}{2} N \overline{mc^2} = \frac{3}{2} NkT$ (1) Justification that $R = kN_A$ (1) with justification (1)	4			4	2	